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Options and Futures

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Fall Term 2025

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The Black-Scholes Model

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- Assumptions of the Black-Scholes model
- Black-Scholes no-arbitrage PDE
- Black-Scholes formula
- Pricing formula vs. pricing bounds
- Black-Scholes as the limit of the binomial model
- Extension: Black-Scholes with dividends

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Black-Scholes assumptions

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- The price of the underlying security (which we call a "stock") follows a geometric Brownian motion.
- The stock pays no dividends over the interval of time under consideration (we will relax this assumption later).
- There is also a money market account (MMA): borrowing and lending at a constant risk-free rate.
- Trading takes place continuously.
- No transaction costs, no indivisibility, no short-sales constraints, no taxes on financial gains.

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Black-Scholes assumptions (cont'd)

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- Process for the stock price is a GBM:

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

  where μ and σ are constant.

- Process followed by the amount deposited in the MMA:

$$dB_t = rB_t dt,$$

  where r is the constant continuously compounded risk-free rate.
  - If you put B_0 in the MMA at time 0, you end up with $B_t = e^{rt}B_0$ at time t.

Black-Scholes no-arbitrage PDE

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- Consider a derivative which pays off at date T an amount contingent on the realization of S_T, say $g(S_T)$.
- Examples:
  - European call option: $g(S_T) = (S_T - K)^+$.
  - European put option: $g(S_T) = (K - S_T)^+$.
  - Forward contract: $g(S_T) = S_T - K$.
- **Problem:** Given S_t, μ, σ, r, T, and $g(\cdot)$, what is the price f_t of the derivative?
- Conjecture: There exists a function F such that the time t price of the derivative can be written as $f_t = F(S_t, t)$.
  - The problem is to pin down the pricing function F.
- We will characterize F as a solution to a particular partial differential equation (PDE).

Continuous delta-hedging

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- Consider the following trading strategy: Sell 1 unit of the derivative, buy δ_t shares of the stock, and invest the proceeds in the MMA, i.e. the amount invested is $\beta_t = f_t - \delta_t S_t$.
- This is a zero-value strategy at time t: the total value (or cost) at t is

$$v_t = -f_t + \delta_t S_t + \beta_t = 0.$$

- δ_t represents a dynamic trading strategy in the underlying over $[0, T]$, with the position in the underlying continuously rebalanced over time.
- δ_t is contingent on the information available at time t (in particular S_t), and may be positive or negative.
- The trick is to choose δ_t such that the return on the portfolio at time t is instantaneously riskless. This is called a continuously delta-hedged portfolio or trading strategy.

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Deriving the BS PDE

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- Since $v_t = -f_t + \delta_t S_t + \beta_t$, we have

$$dv_t = -df_t + \delta_t dS_t + d\beta_t.$$

- Applying Itô's lemma to $f_t = F(S_t, t)$, we get

$$\begin{aligned} df_t &= \frac{\partial F}{\partial t} dt + \frac{\partial F}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 F}{\partial S^2} (dS_t)^2 \\ &= \left(\frac{\partial F}{\partial t} + \frac{1}{2} \frac{\partial^2 F}{\partial S^2} \sigma^2 S_t^2 \right) dt + \frac{\partial F}{\partial S} dS_t. \end{aligned}$$

- Throughout this argument, the partial derivatives of F are evaluated at (S_t, t).
- We can make the strategy instantaneously riskless by setting $\delta_t = \frac{\partial F}{\partial S}$, evaluated at (S_t, t).

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Deriving the BS PDE (cont'd)

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- For this choice of δ_t, dv_t has no dependence on dS_t:

$$dv_t = -\left(\frac{\partial F}{\partial t} + \frac{1}{2}\frac{\partial^2 F}{\partial S^2}\sigma^2 S_t^2\right) dt + d\beta_t.$$

- Moreover,

$$d\beta_t = r\beta_t dt = r\left(f_t - \delta_t S_t\right) dt = r\left(F(S_t, t) - \frac{\partial F}{\partial S} S_t\right) dt.$$

- Therefore, dv_t is given by

$$-\left(\frac{\partial F}{\partial t} + r\frac{\partial F}{\partial S} S_t + \frac{1}{2}\frac{\partial^2 F}{\partial S^2}\sigma^2 S_t^2 - rF(S_t, t)\right) dt.$$

- But $v_t = 0 \Rightarrow dv_t = 0$ (by no arbitrage), i.e.,

$$\frac{\partial F}{\partial t} + r\frac{\partial F}{\partial S} S_t + \frac{1}{2}\frac{\partial^2 F}{\partial S^2}\sigma^2 S_t^2 - rF(S_t, t) = 0.$$

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The Black-Scholes PDE

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- Using subscripts to denote the partial derivatives of F and dropping time subscripts, we can write the BS PDE as follows:

$$F_t(S,t) + rSF_s(S,t) + \frac{1}{2}\sigma^2 S^2 F_{ss}(S,t) - rF(S,t) = 0$$

- In addition, no arbitrage implies $F(S,T) = g(S)$ for all S.
- Thus pricing a derivative in the Black-Scholes model amounts to solving a PDE given some terminal condition.
- In some cases, it is possible to guess a closed-form solution. Otherwise, the PDE can be solved numerically.
- Remark: μ does not enter anywhere in the PDE, i.e., the rate at which the expected stock price grows does not affect the no-arbitrage price of the derivative (cf. binomial model).

Feynman-Kac theorem

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- An alternative way to solve the BS PDE is to use Feynman-Kac:
  Theorem (Feynman-Kac): F is the solution of the BS PDE with terminal condition $F(S,T) = g(S)$ if and only if

$$F(S,t) = e^{-r(T-t)}\mathbb{E}[g(S_T)],$$

  where the expectation is computed under the assumption that

$$dS_t = rS_t dt + \sigma S_t dW_t,$$

  with initial condition $S_t = S$.
- The Feynman-Kac formula for F is in fact the same as the risk-neutral valuation formula that we will see later: in the absence of arbitrage there is a risk-neutral probability Q, such that:
  - the dynamics of the stock under Q are given by (1);
  - the no-arbitrage price of a derivative is equal to its expected payoff under Q discounted at the risk-free rate.

Monte-Carlo pricing in Black-Scholes

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In practice, the Feynman-Kac formula suggests a way to numerically compute the no-arbitrage price $f_t = F(S_t, t)$ by Monte-Carlo simulation as follows:

- Simulate a sample path of $dS_t = rS_t dt + \sigma S_t dW_t$ over $[t, T]$ starting at S_t:
  - This is done by taking independent draws from a standard normal distribution and successively calculating values of S_t satisfying $\Delta S_t = rS_t \Delta t + \sigma S_t \Delta W_t$.
  - Note that ΔW_t has the same distribution as $(\sqrt{\Delta t})z$, where z is standard normal.
- Simulate many such sample paths.
- Compute $g(S_T)$ for each sample path.
- Compute the average of $g(S_T)$ over all sample paths.
- Discount this average at the risk-free rate.

Black-Scholes formula for a European call option

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- In the case of standard European call and put options, it turns out that the price of the derivative can be obtained in closed form.
- Consider a European call option with payoff function $g(S_T) = (S_T - K)^+$.
- Let $C(S, t)$ denote the call price function, with the time t price given by $C(S_t, t)$.
- Using Feynman-Kac, this price can be obtained as follows:

$$C(S, t) = e^{-r(T-t)}\mathbb{E}_t[(S_T - K)^+],$$

under the assumption $dS_t = rS_t dt + \sigma S_t dW_t$, and initial condition $S_t = S$.

- Since the stock price is a geometric Brownian motion, S_T is lognormally distributed.

$\mathbb{E}[(X-K)^+]$ for X lognormal

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- **Result:** Suppose X is lognormally distributed, and $\mathrm{Var}(\log X) = \sigma^2(T-t)$. Then

$$\mathbb{E}[(X-K)^+] = \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2),$$

where Φ is the standard normal cdf and

$$d_1 = \frac{\log\frac{\mathbb{E}(X)}{K} + \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}.$$

- Proof: See Appendix A.

Deriving the Black-Scholes Formula

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- If S follows $dS_t = rS_t dt + \sigma S_t dW_t$, then

$$\log S_T \sim N\left(\log S_t + (r - \frac{1}{2}\sigma^2)(T-t), \sigma^2(T-t)\right),$$

i.e., S_T is lognormal with $\mathbb{E}_t(S_T) = e^{r(T-t)}S_t$ and $\mathrm{Var}(\log S_T) = \sigma^2(T-t)$.

- Therefore,

$$\mathbb{E}_t[(S_T - K)^+] = e^{r(T-t)}S_t\Phi(d_1) - K\Phi(d_2),$$

where

$$d_1 = \frac{\log(S_t/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}$$

and

$$d_2 = d_1 - \sigma\sqrt{T-t}.$$

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Black-Scholes Formula

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Result:
In the Black-Scholes model, the price of a European call option at $t < T$ is given by

$$C(S,t) = S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2),$$

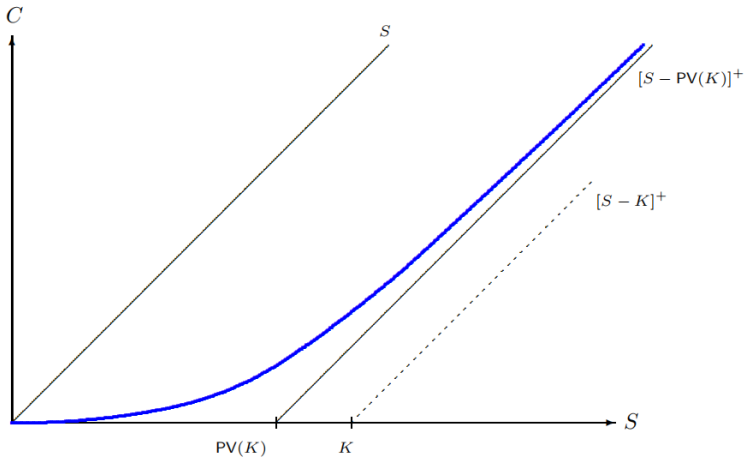
where $\Phi(\cdot)$ is the standard normal cdf, and

$$d_1 = \frac{\log(S/K) + (r + \frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}},$$

$$d_2 = d_1 - \sigma\sqrt{T-t}.$$

The Black-Scholes Call Price $C(\cdot, t)$

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Comments on the BS Formula

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- The BS formula for the call price is

$$C(S,t) = S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2).$$

- The call price depends on the variables S and t, and the parameters σ, r, T, K. It does not depend on μ.
- The call can be replicated at t by buying $\Phi(d_1)$ shares, and borrowing the amount $e^{-r(T-t)}K\Phi(d_2)$ in the money market.
- This portfolio needs to be continuously rebalanced over time. The right hand side of the BS formula is the initial cost of the (self-financing) replicating strategy.
- A call option is essentially a dynamic leveraged position in the underlying. The two terms in the BS formula reflect respectively the value of the risky and riskfree holdings in the replicating strategy.

Delta-hedging interpretation

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- From the BS formula for the call price, we have

$$-C_t + \underbrace{\Phi(d_1)}_{\delta_t} S_t + \underbrace{\left[-e^{-r(T-t)}\right] K\Phi(d_2)}_{\beta_t} = 0.$$

- Recall that a zero-value delta-hedged portfolio is constructed by going short in the call, buying δ_t shares of the stock, and investing the proceeds β_t in the money market account.
- We will show later that

$$\delta_t = \frac{\partial C}{\partial S} = \Phi(d_1).$$

- For a call option, $\beta_t < 0$.

Call price

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- The call price C is increasing and convex in S.
- Asymptotically C is linear in S:

$$\lim_{S\to\infty} C = S - \mathrm{PV}(K).$$

- Intuition:The call option will be exercised for sure, so the option feature itself has no value
  - The value of the option is just its intrinsic value
  - This is equal to the value of a forward contract with payoff $S_T - K$.

Black-Scholes formula for a European put option

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- Denote the put price function by $P(S,t)$, with the time t put price given by $P(S_t,t)$.
- Using put-call parity,

$$\begin{aligned} P(S,t) &= C(S,t) + e^{-r(T-t)}K - S \\ &= S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2) + e^{-r(T-t)}K - S \\ &= -S[1-\Phi(d_1)] + e^{-r(T-t)}K[1-\Phi(d_2)]. \end{aligned}$$

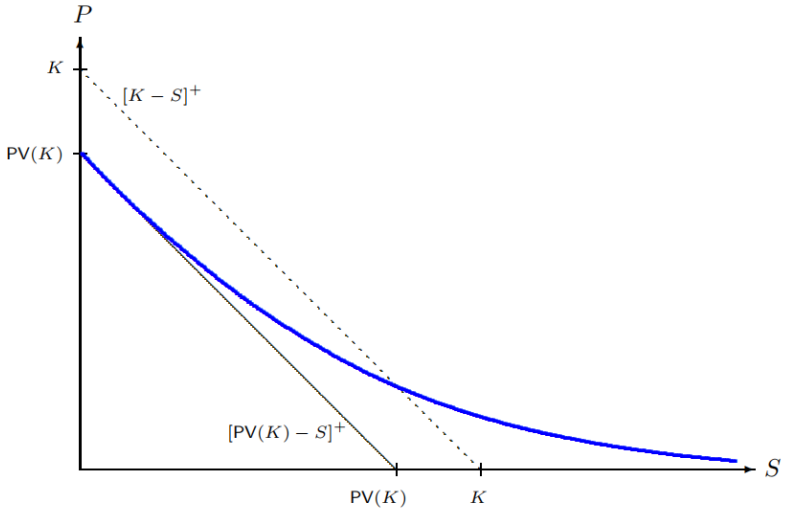
- Hence,

$$P(S,t) = -S\Phi(-d_1) + e^{-r(T-t)}K\Phi(-d_2).$$

- A put option is replicated by shorting the underlying and lending in the money market.

The Black-Scholes put price $P(\Delta, t)$

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Put price

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- The put price P is decreasing and convex in S.
- Also

$$\lim_{S\to 0} P = \mathrm{PV}(K)$$

$$\lim_{S\to \infty} P = 0$$

- Note: Whereas C' always remains above $(S-K)^+$, P is below $(K-S)^+$ for S lower than some threshold.

BS pricing formula vs pricing bounds

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- Recall the no-arbitrage bounds on the prices of European call and put options:

$$[S - \mathrm{PV}(K)]^+ \leq C \leq S$$

$$[\mathrm{PV}(K) - S]^+ \leq P \leq \mathrm{PV}(K)$$

- These pricing bounds do not rely on any distributional assumption on the underlying process; they are robust to any underlying dynamics.
- By contrast, the restrictive distributional assumptions made in the BS model have allowed us to go beyond pricing bounds and to pin down a unique no-arbitrage price.
- Exercise: Check that call and put prices given by the BS formula are inside the bounds.

Black-Scholes as the limit of the binomial model

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- Consider an n-step binomial tree from time 0 to T, with each iteration of length $\Delta t = \frac{T}{n}$.
- As $n \to \infty$ (and $\Delta t \to 0$),
  - the binomial process for the underlying price converges to a geometric Brownian motion,
  - the n-step binomial option price formula converges to the Black-Scholes formula.
- Binomial trees are often used for numerical computation.

Calibrating a binomial tree

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- We want to choose the parameters of the binomial process for S so that in the limit S follows a GBM with drift μ and volatility σ, i.e.,

$$\log S_t = \log S_0 + \hat{\mu} t + \sigma W_t,$$

where $\hat{\mu} = \mu - \frac{\sigma^2}{2}$.

- Choose u and d to reflect estimates of $\hat{\mu}$ and σ. Proposed by Cox, Ross and Rubinstein:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}, \quad p = \frac{1}{2} + \frac{1}{2}\left(\frac{\hat{\mu}}{\sigma}\right)\sqrt{\Delta t}.$$

- For these values, we can verify that:

$$\mathbb{E}[\log(S_t/S_0)] = \hat{\mu} t, \quad \mathrm{Var}[\log(S_t/S_0)] = (\sigma^2 - \hat{\mu}^2 \Delta t) t.$$

- The risk-neutral probability of an up move is

$$q = \frac{e^{r\Delta t} - d}{u - d}.$$

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BS when the underlying pays known lumpy dividends

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- Suppose you are at time t. The underlying asset is known to pay a dividend D at time τ, $t < \tau < T$, and the current price of the underlying is S.
- In addition to the replication strategy for the no-dividend case, you can borrow $e^{-r(\tau-t)}D$ for each share bought, with repayment at time τ. Then the cost per share is $S - e^{-r(\tau-t)}D$.
- At τ, you break even; you receive D per share and repay your loan.
- Since the replicating strategy is self-financing, we just need to consider the cost at time t.
- The BS formulas need to be amended by plugging $S - e^{-r(\tau-t)}D$ in place of S. For instance, the price of a call option is:

$$C(S,t) = \left(S - e^{-r(\tau-t)}D\right)\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2),$$

with

$$d_{1,2} = \frac{\log\left(\frac{S-e^{-r(\tau-t)}D}{K}\right) + \left(r \pm \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}.$$

The case of a constant dividend yield

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- Let D_t be the cumulative dividends paid between 0 and t. We say that the underlying pays a constant dividend yield q if

$$dD_t = qS_t\,dt.$$

- Let H_t denote your holdings of the underlying asset at time t. If you reinvest the instantaneous dividend in the asset, your holdings of the asset increase by

$$dH_t = \frac{H_t \times qS_t\,dt}{S_t} = qH_t\,dt.$$

- Therefore, if you start with H_t and keep reinvesting the dividends between times t and T, you end up with $H_T = e^{q(T-t)}H_t$.
- For each share that you would have bought at t in the no-dividend case, you buy instead $e^{-q(T-t)}$ shares, and keep reinvesting the dividend until T. This holding then grows to one share at T.

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BS when the underlying pays a constant dividend yield

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- The BS formulas need to be amended by plugging $e^{-q(T-t)}S$ in place of S.
- For instance, the price of a call option is

$$C(S,t) = e^{-q(T-t)}S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2),$$

with

$$d_{1,2} = \frac{\log\left(\frac{S}{K}\right) + \left(r - q \pm \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}.$$

- This is relevant for the pricing of index and currency options.

Example: Currency options

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- Consider an option on a foreign currency
  - Say the price of one unit of foreign currency in domestic currency is given by the exchange rate S. This is the underlying of the option.
- Holding foreign currency allows us to save at the foreign riskfree rate r_F. These interest payments are akin to a constant dividend yield $q = r_F$.
- Therefore, if S is the current exchange rate, the price of an option on that currency with strike K can be written as above, substituting r_F for q.
- Notice that the price of a currency option involves two interest rates – the domestic interest rate r and the foreign interest rate r_F.

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- Appendix A contains proofs of some results
- This material will not be examined

$\mathbb{E}[(X-K)^+]$ for X lognormal

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- To derive the Black-Scholes formula, we used the following result: Suppose X is lognormally distributed, and $\text{Var}(\log X) = \sigma^2(T-t)$. Then

$$\mathbb{E}[(X-K)^+] = \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2),$$

where Φ is the standard normal cdf and

$$d_1 : \frac{\log[\mathbb{E}(X)/K] + \sigma^2(T-t)/2}{\sigma\sqrt{T-t}}, \quad d_2 := d_1 - \sigma\sqrt{T-t}$$

- To simplify notation, let $\mathbb{E}(\log X) = \alpha$, and $\text{Var}(\log X) = \beta^2$ (i.e. $\beta := \sigma\sqrt{T-t}$)

Proof

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- We can write $\log X = \alpha + \beta z$, where z is standard normal. Thus $X = e^{\alpha+\beta z}$ and $\mathbb{E}(X) = e^{\alpha+\beta^2/2}$. Moreover, $X \geq K$ is equivalent to $\log X \geq \log K$,or

$$\begin{aligned} z &\geq \frac{\log K - \alpha}{\beta} \\ &= \frac{\log K - \log \mathbb{E}(X) + \beta^2/2}{\beta} \\ &= -\frac{\log[\mathbb{E}(X)/K] - \beta^2/2}{\beta} \\ &= -d_2. \end{aligned}$$

- Recall the standard normal density: $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$
- Let Φ be the standard normal cdf.

Proof(cont'd)

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Thus we have:

$$\begin{aligned}
\mathbb{E}[(X-K)^+] &= \int_{-d_2}^{\infty} (e^{\alpha+\beta z} - K)\phi(z)dz \\
&= \int_{-d_2}^{\infty} e^{\alpha+\beta z}\phi(z)dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \int_{-d_2}^{\infty} \mathbb{E}(X)e^{-\frac{1}{2}\beta^2+\beta z}\cdot\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}z^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}dz - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_2-\beta}^{\infty}\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(z-\beta)^2}d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\int_{-d_1}^{\infty}\phi(z-\beta)d(z-\beta) - K\int_{-d_2}^{\infty}\phi(z)dz \\
&= \mathbb{E}(X)\Phi(d_1) - K\Phi(d_2).
\end{aligned}$$

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Breeden-Litzenberger

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- Let $\varphi(\cdot)$ be the density of S_T under the risk-neutral probability Q The price of a call option at $t = 0$ as a function of the strike price K can be written as:

$$C(K) = e^{-rT}\mathbb{E}^Q[(S_T - K)^+] = e^{-rT}\int_K^\infty (x - K)\varphi(x)dx$$

- Result [Breeden-Litzenberger (1978)]:

$$\varphi(K) = e^{rT}C''(K).$$

Proof of Breeden-Litzenberger result

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Proof: We can write

$$e^{rT}C(K) = \int_K^\infty x\varphi(x)dx - K\int_K^\infty \varphi(x)dx.$$

Differentiating:

$$\begin{aligned} e^{rT}C'(K) &= \frac{d}{dK}\int_K^\infty x\varphi(x)dx - K\frac{d}{dK}\int_K^\infty \varphi(x)dx - \int_K^\infty \varphi \\ &= -K\varphi(K) - K[-\varphi(K)] - \int_K^\infty \varphi(x)dx \\ &= -\int_K^\infty \varphi(x)dx. \end{aligned}$$

Differentiating again:

$$e^{rT}C''(K) = -[-\varphi(K)] = \varphi(K).$$

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- Put-call parity (non-dividend paying stock):

$$C_t = P_t + S_t - e^{-r(T-t)}K$$

- Lognormal property of GBM:
  Suppose $dS_t = \mu S_t dt + \sigma S_t dW_t$. Then the conditional distribution of $\log S_T$ at time t is normal:

$$\log S_T \sim \mathcal{N}\left[\log S_t + \left(\mu - \sigma^2/2\right)(T-t), \sigma^2(T-t)\right]$$

- Black-Scholes formula:
  The price of a European call option at $t < T$ is given by

$$C(S,t) = S\Phi(d_1) - e^{-r(T-t)}K\Phi(d_2)$$

  where $\Phi(\cdot)$ is the standard normal cdf, and

$$d_1 = \frac{\log(S/K) + \left(r + \sigma^2/2\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

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- Black-Scholes PDE:
  Consider a derivative on a stock with payoff $g(S_T)$ at time T, whose price at time $t \leq T$ is given by $F(S,t)$. The Black-Scholes PDE is

$$F_t(S,t) + rSF_S(S,t) + \frac{1}{2}\sigma^2 S^2 F_{SS}(S,t) - rF(S,t) = 0$$

- Standard normal density:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- Expectation of the exponential of a normal random variable:
  If X normally distributed, then

$$\mathbb{E}[\exp(X)] = \exp\left[\mathbb{E}(X) + \frac{1}{2}\operatorname{Var}(X)\right]$$

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